Early Warning Prediction of Coral Bleaching Risk in the Central Saudi Arabian Red Sea Using Machine Learning

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Abstract

Coral bleaching in the central Saudi Arabian Red Sea is becoming more frequent and more severe. Field surveys recorded bleaching in 65% of coral colonies in October 2020, and in 90% of coral cover at 5 m depth in September 2024. The region has limited long-term monitoring, and existing global forecasts are not tuned to local conditions. This study tested whether a machine learning model trained on public satellite data could forecast bleaching risk several weeks ahead.

Daily sea surface temperature records from the NOAA Coral Reef Watch virtual station for the Al Madinah and Makkah region were combined with the Indian Ocean Dipole Mode Index and satellite solar radiation data. The dataset covered March 2004 to December 2022. Thermal stress in this region occurs only between August and December, so model development was limited to those months. An XGBoost classifier was trained to separate three categories: no risk, urgent risk within two weeks, and distant risk within three to six weeks. Two models were compared. The first used three thermal variables. The second added three supplementary environmental predictors. Two simpler baselines were also included.

On test data from 2019 to 2022, the baseline model separated urgent risk from no risk with an F1 score of 0.972, outperforming both simpler baselines. The supplementary predictors did not improve performance. For the 2020 event, the model issued its first warning ten days before the observed Degree Heating Week crossed the bleaching threshold. For the 2024 event, which fell outside both the training and testing periods, the lead time was again ten days, at a nearly identical thermal value and confidence level. Performance was weak beyond two weeks, since few training examples were available for longer lead times.

A localized early-warning system for this region can therefore be built using data that are already public.

1. Introduction

Coral reefs cover less than 1% of the seafloor, yet they support at least 25% of all marine species (UNEP, 2026). They also protect coastlines by reducing wave energy by an average of 97% (Ferrario et al., 2014), and they provide economic value through fisheries and tourism. Despite this importance, coral reefs worldwide are declining rapidly, mainly because of rising sea surface temperatures that cause coral bleaching.

Bleaching happens when corals are exposed to high temperatures and expel the algae living inside them. This causes corals to lose their color and their main source of energy. Corals can recover if temperatures return to normal quickly, but prolonged heat stress often leads to coral death, which affects the whole reef structure.

The Red Sea is a useful place to study this problem. It is a narrow, semi-enclosed sea where surface temperatures have risen by up to 0.45°C per decade over the last 30 years. This warming has increased the frequency and severity of bleaching events. In October 2020, a major bleaching event occurred along the central Saudi Arabian coast. Surveys found bleaching in 65% of coral colonies, and almost half of them eventually died (Dunn et al., 2025). A more severe event followed in September 2024. At 5 m depth, 90% of coral cover showed bleaching, and coral cover later declined by 65%, with several key taxa nearly

disappearing (Dunn et al., 2026). These events are becoming more frequent and more severe, which makes early-warning tools important for this region.

However, not all parts of the Red Sea are affected equally. Osman et al. (2018) compared bleaching records with sea surface temperature data from 1982 to 2012 and found that the northern Red Sea did not experience mass bleaching, even during long periods of high heat. This was different from the central and southern Red Sea, which led researchers to suggest that the northern region acts as a natural thermal refuge. In contrast, corals in the central Red Sea live much closer to their upper temperature limits, since the highest water temperatures in this area are close to the bleaching threshold (Osman et al., 2018). This means corals in the central region have less room before heat stress leads to bleaching. The events in 2015 near Thuwal, in 2020, and in 2024 all confirm this vulnerability, which is why the central Red Sea is the focus of this study. At the same time, the Red Sea remains a data-scarce region compared to other well-studied reef systems.

Coral reef monitoring worldwide relies mainly on NOAA's Coral Reef Watch program. It uses satellite sea surface temperature data and the Degree Heating Week (DHW) metric to track accumulated thermal stress. The program also issues a Four-Month Coral Bleaching Outlook, a weekly forecast based on the CFSv2 climate model at 0.5 degree resolution. These products work well at a global scale. However, their accuracy varies between regions, and their resolution is coarse compared to individual reef sites.

Machine learning has been applied to coral reef research in several ways, including image-based detection of bleaching and modeling of reef connectivity. However, few studies have developed localized systems that predict when bleaching is likely to begin. Arcodia et al. (2025) addressed this gap for Florida's Coral Reef, using an XGBoost model combined with explainable AI to forecast heat stress onset up to six weeks in advance, outperforming simpler baseline models. This type of localized approach has not yet been applied to the Red Sea, despite the region's documented vulnerability and lack of long-term data. Other studies have also suggested that predictors beyond DHW, such as climate indices and historical thermal variability, may improve prediction accuracy, although this has not been tested in other reef regions.

This study addresses this gap by developing a machine learning framework to predict coral bleaching risk in the central Saudi Arabian Red Sea, based on a method previously applied in a different environmental context. The study has three main objectives: first, to evaluate whether a localized XGBoost model can produce useful predictions of bleaching risk using publicly available sea surface temperature data; second, to test whether adding regional climate and radiation indices improves prediction accuracy in this data-scarce region; and third, to validate the model against two independently documented bleaching events — the 2020 event and the more severe 2024 event — to determine whether it would have provided a useful early-warning period in each case.

2. Data and Methods

2.1 Study Area and Data Sources

This study focuses on the central Saudi Arabian Red Sea. Sea surface temperature data were obtained from the NOAA Coral Reef Watch Regional Virtual Station for the Al Madinah and Makkah region, centered at 21.125°N and 39.45°E. This station covers the coastal area between Rabigh and Yanbu, where the 2020 and 2024 bleaching events were documented.

Three data sources were used. The first was NOAA Coral Reef Watch, which provides daily records from 1985 onward. Three variables were extracted: the Degree Heating Week (DHW), which measures accumulated thermal stress over the previous 12 weeks; the HotSpot value, which measures instantaneous

heat stress above the monthly maximum mean; and the sea surface temperature anomaly. The full record contained 15,214 daily observations from January 1985 to August 2026.

The second source was the Dipole Mode Index (DMI) from NOAA's Physical Sciences Laboratory, which represents the Indian Ocean Dipole. This index was chosen because the Red Sea is more closely connected to Indian Ocean climate patterns than to the Pacific systems used in other regions. After removing missing values, 1,877 monthly records were available from January 1870 to May 2026.

The third source was the CAMS Solar Radiation Time-Series from the Copernicus Atmosphere Data Store, which provides daily global horizontal irradiance (GHI) under both clear-sky and actual weather conditions. A cloud ratio was calculated as actual GHI divided by clear-sky GHI, where values close to 1 indicate clear skies and lower values indicate greater cloud cover. This dataset contained 6,902 daily records from February 2004 to December 2022.

2.2 Data Processing

The three sources were merged by date. Since the DMI is reported monthly, each daily record was matched to its corresponding monthly value. The merged dataset was limited by the shortest record, the solar radiation data, which begins in February 2004. Merging produced 6,902 daily records with no missing values.

An additional predictor, historical thermal variability (histSSTSD), was calculated as the standard deviation of mean daily sea surface temperature over the preceding 30 days. This variable indicates whether temperatures in the recent past were stable or fluctuating. Because this calculation requires 30 prior days, the first 29 records were removed, leaving a final dataset of 6,873 daily records from 1 March 2004 to 31 December 2022.

2.3 Defining the Prediction Target

Monthly analysis of the DHW record showed that thermal stress in this region follows a clear seasonal pattern (Figure 1). From February to May, DHW remained at zero across all study years. Values began rising in August (mean 1.69), continued through September (mean 3.46), and peaked in October (mean 4.40, maximum 17.42), before declining through November and December.

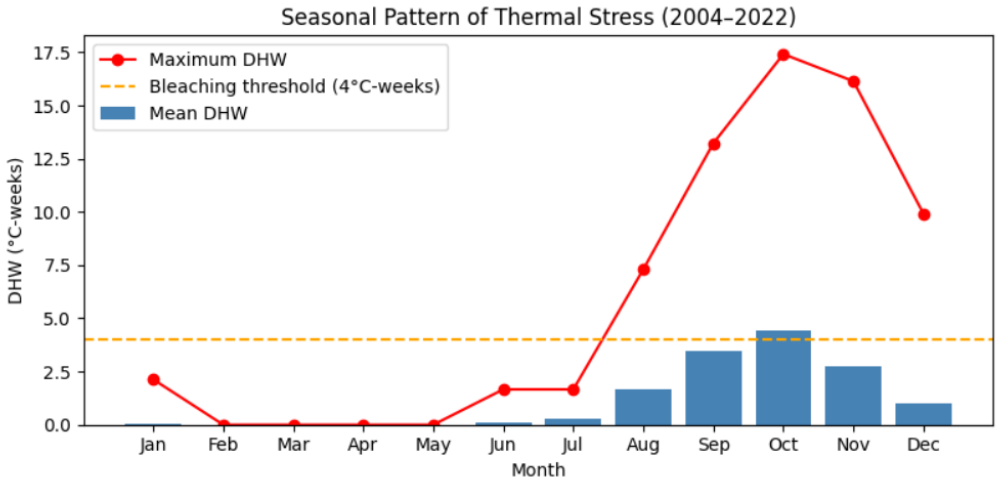


Figure 1. Seasonal pattern of thermal stress in the study area (2004–2022). Bars show mean DHW per month; the red line shows maximum recorded DHW; the dashed line marks the 4 °C-weeks bleaching threshold.

Based on this pattern, only records from August to December were used for model development, giving 2,904 records across 19 seasons.

Following the NOAA threshold, the onset of moderate heat stress was defined as the point at which DHW reaches 4.0 °C-weeks. For each day within the season, the number of weeks remaining until this threshold was calculated, with a maximum lead time of six weeks.

Initial modeling used eight separate classes (0 to 6 weeks, plus a no-onset class). However, the class distribution was highly uneven: 1,970 no-onset records and 756 records at zero weeks, compared with only 14, 10, and 7 records at four, five, and six weeks respectively. This imbalance produced poor performance for the longer lead times. The target was therefore simplified into three categories: no risk within six weeks, urgent risk (onset within 0–2 weeks), and distant risk (onset within 3–6 weeks).

2.4 Model Development

Training and testing data were separated by time rather than randomly, since random splitting of time-series data allows information from adjacent days to leak between the two sets. Records from 2004 to 2018 were used for training (2,292 days), and records from 2019 to 2022 were reserved for testing (612 days). The 2020 season was retained in the test set so that predictions could be compared against the independently documented bleaching event.

An XGBoost classifier was used, following the approach of Arcodia et al. (2025). Tree-based methods are better suited than deep learning architectures for datasets of this size, and their structure allows model decisions to be examined using explainability methods.

The class distribution in the training data remained uneven after simplification, with 1,719 no-risk days, 516 urgent-risk days, and 57 distant-risk days. Balanced class weights were therefore applied during training to give greater weight to the less frequent categories.

Two models were trained for comparison. The baseline model used only the three thermal variables (DHW, HotSpot, and SST anomaly). The extended model added the three supplementary predictors (histSSTSD, DMI, and cloud ratio) to test whether additional environmental information improves prediction accuracy in this region.

Two simpler baselines were also included for comparison, following the approach of Arcodia et al. (2025). The first was a multinomial logistic regression using the same three thermal variables and the same class weights. The second was a frequency model that assigned to each day of the year the class most commonly observed on that day in the training data, using no environmental variables at all.

2.5 Model Evaluation

Model performance was assessed using precision, recall, and F1 score for each category. The F1 score was used as the primary measure, since accuracy alone can be misleading when classes are unbalanced.

To identify which variables influenced the model's decisions, SHAP (SHapley Additive exPlanations) values were calculated on the test data. This method quantifies how much each predictor contributes to each prediction, allowing feature importance to be compared across the three risk categories.

Finally, the model was applied to two documented bleaching events. The first was the 2020 event, which falls within the test period. The second was the 2024 event, which lies entirely outside both the training and testing periods and therefore provides an independent external validation. Because the solar radiation dataset ends in December 2022, only the baseline model could be applied to the 2024 event.

3. Results

3.1 Model Performance

All models were tested on the 2019–2022 data (612 days). The baseline model, which used only the three thermal variables, reached an overall accuracy of 0.933. Performance differed across the three risk categories (Figure 2). The urgent-risk category, which matters most for early warning, reached an F1 score of 0.972, with precision of 0.988 and recall of 0.957. The no-risk category reached 0.931. The distant-risk category performed poorly, with an F1 score of 0.125.

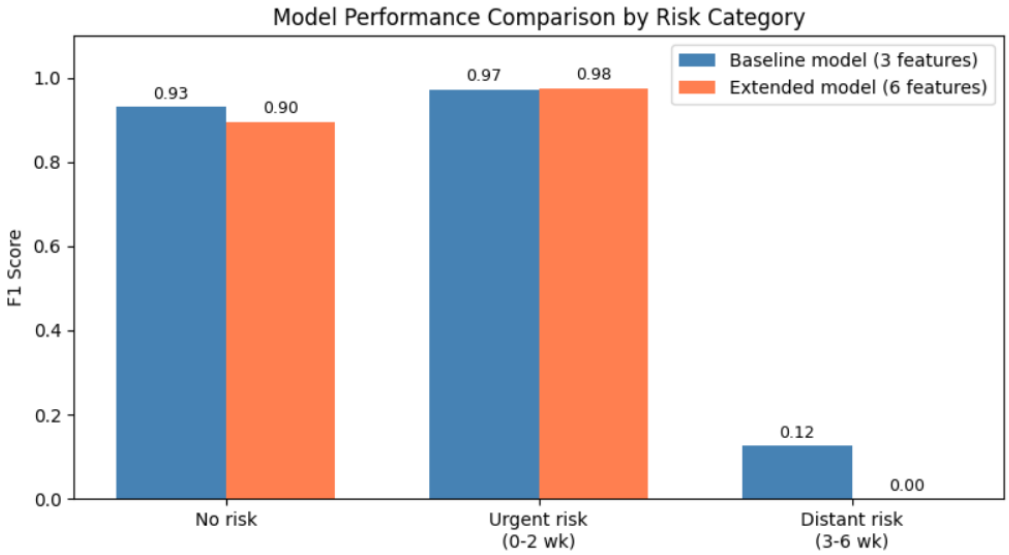


Figure 2. F1 score by risk category for the baseline model (three thermal variables) and the extended model (six variables), evaluated on the 2019–2022 test set.

The extended model added historical thermal variability, the Dipole Mode Index, and cloud ratio. It did not improve performance. Its accuracy was 0.913, slightly below the baseline. The urgent-risk category gained very little (0.975 compared with 0.972), but the no-risk category dropped from 0.931 to 0.895, and the distant-risk category fell to zero, meaning the model did not identify any case in this category.

Both simpler baselines performed worse overall (Table 1). The frequency model, which used calendar position alone, reached an accuracy of 0.410 and failed entirely on both risk categories, predicting no risk on every test day. This confirms that seasonal timing alone cannot identify when thermal stress will begin. Logistic regression performed considerably better, with an accuracy of 0.879 and an F1 score of 0.953 for urgent risk, but remained below the baseline XGBoost model in both measures. One exception is worth noting: for the distant-risk category, logistic regression achieved an F1 score of 0.241, higher than the 0.125 obtained by XGBoost. The simpler model appears less prone to overfitting on a category represented by only 57 training days, although both scores remain too low for practical use.

Model	No risk	Urgent risk	Distant risk	Accuracy
XGBoost (baseline)	0.931	0.972	0.125	0.933
XGBoost (extended)	0.895	0.975	0.000	0.913
Logistic regression	0.883	0.953	0.241	0.879
Frequency model	0.582	0.000	0.000	0.410

Table 1. Model performance on the 2019–2022 test set. Values are F1 scores per category.

These results suggest that the three thermal variables already carried the predictive information needed here, that the extra environmental predictors did not add useful signal with the data available, and that the relationship between thermal variables and bleaching risk is not linear.

3.2 Feature Importance

SHAP values were calculated on the test data to see which variables drove the baseline model's decisions (Figure 3). The ranking of the three predictors changed depending on the risk category.

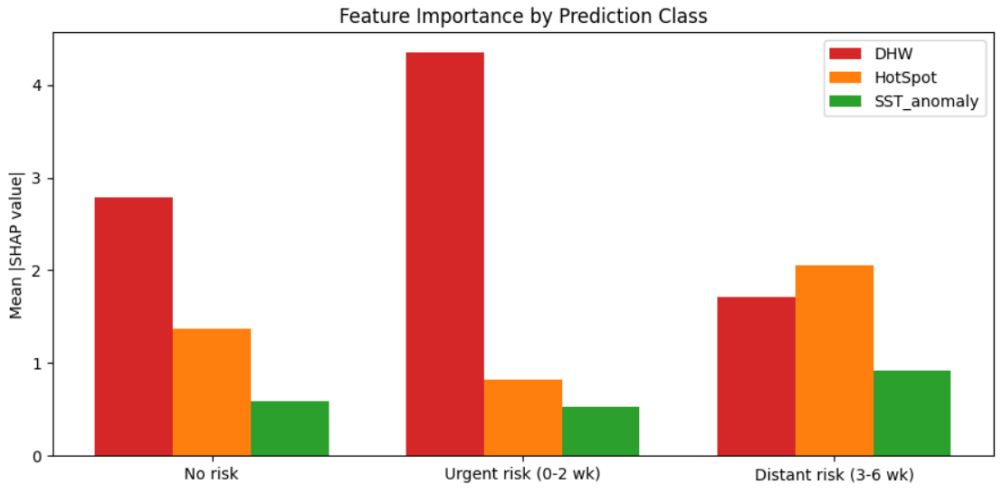


Figure 3. Mean absolute SHAP value for each predictor, shown separately for the three risk categories.

For urgent-risk predictions, DHW was clearly dominant, with a mean absolute SHAP value of 4.349, followed by HotSpot at 0.827 and SST anomaly at 0.526. The same order appeared for no-risk predictions, where DHW ranked first at 2.788, then HotSpot at 1.370 and SST anomaly at 0.594.

The distant-risk category showed a different pattern. HotSpot became the strongest predictor at 2.056, ahead of DHW at 1.715, with SST anomaly at 0.916. This suggests that for longer lead times the model depended more on the instantaneous heat signal than on accumulated stress, which fits the fact that accumulated stress has not yet built up at that point in the season.

3.3 Validation Against the 2020 Bleaching Event

The model was applied to the 2020 season, which falls inside the test period. The first urgent-risk warning was issued on 11 August 2020, when DHW was 1.79, with a predicted probability of 0.953. The observed DHW crossed the 4.0 °C-weeks threshold ten days later, on 21 August 2020, and kept rising to a seasonal peak of 17.42 °C-weeks.

The field surveys that recorded bleaching in 65% of coral colonies were carried out in October 2020 (Dunn et al., 2025). The first warning therefore came about eight weeks before the documented survey period. The source study does not report the exact date when bleaching began, so the lead time is measured against the survey period rather than against bleaching onset.

3.4 External Validation Against the 2024 Bleaching Event

The 2024 season falls outside both the training period (2004–2018) and the test period (2019–2022), so it provides an independent check. Because the solar radiation data ends in December 2022, only the baseline model could be applied.

The first urgent-risk warning for 2024 was issued on 17 June 2024, when DHW was 1.88, with a predicted probability of 0.974. The observed DHW crossed the threshold ten days later, on 27 June 2024, and reached a seasonal peak of 22.31 °C-weeks (Figure 4), higher than the 17.42 recorded in 2020.

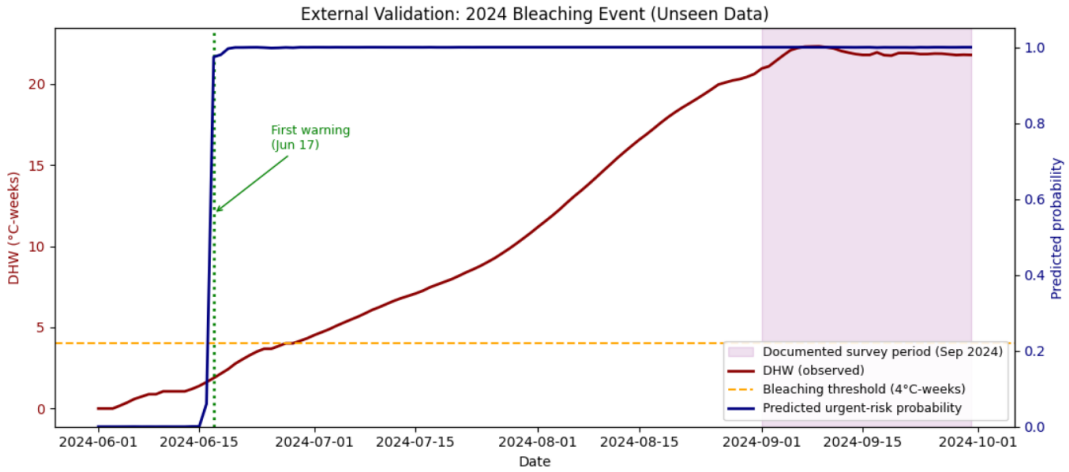


Figure 4. Model output for the 2024 season. The red line shows observed DHW; the blue line shows the predicted urgent-risk probability; the shaded band marks the documented survey period.

The model behaved almost identically across the two events. It issued the warning at a similar DHW value (1.79 in 2020 and 1.88 in 2024), with similar confidence (0.953 and 0.974), and with the same lead time of ten days before the threshold was crossed. Field surveys recorded bleaching in 90% of coral cover at 5 m depth in September 2024 (Dunn et al., 2026), about eleven weeks after the first warning.

4. Discussion

4.1 Early Warning Performance

The model produced useful early warnings in both documented bleaching events. In 2020, the first urgent-risk warning came ten days before DHW crossed the bleaching threshold, and about eight weeks before the field surveys that recorded bleaching. In 2024, the pattern repeated almost exactly: the warning came ten days before the threshold was crossed, and about eleven weeks before the September surveys.

What matters here is not only that the warnings were early, but that they were consistent. The model issued its warning at a DHW value of 1.79 in 2020 and 1.88 in 2024, with confidence levels of 0.953 and 0.974. Since the 2024 season was never seen during training or testing, this consistency suggests the model learned a stable relationship between rising thermal stress and approaching bleaching risk, rather than fitting patterns specific to one period.

The comparison with simpler baselines supports this. A model using calendar position alone failed completely, which shows that seasonal timing does not determine when stress begins. A linear model performed reasonably but remained below the tree-based model, indicating that the relationship between thermal variables and risk is not a simple linear one.

The practical value of this lead time depends on what can be done with it. Ten days before the threshold, and several weeks before visible bleaching, is enough time to plan field surveys, increase monitoring, or prepare protective actions at specific sites.

4.2 The Extended Model Did Not Improve Performance

Adding historical thermal variability, the Dipole Mode Index, and cloud ratio did not improve prediction accuracy. The extended model performed slightly worse overall (0.913 compared with 0.933), and failed completely on the distant-risk category.

This result was not expected. Previous work has suggested that predictors beyond DHW may improve bleaching prediction, and the three added variables were chosen because each has a plausible physical link to thermal stress. Several factors may explain why they did not help here.

First, the added variables may overlap with information already contained in the thermal predictors. Historical thermal variability is calculated from the same sea surface temperature record used for DHW and HotSpot, so it may repeat rather than add signal.

Second, the training set contained 2,292 days. Adding three predictors to a dataset of this size increases model complexity without providing enough examples to learn the new relationships reliably.

Third, the Dipole Mode Index operates on a seasonal to multi-year timescale, while the prediction target here is measured in weeks. A climate index may influence whether a season becomes unusually warm, without helping to time the onset within that season.

This finding has a practical benefit. Because the baseline model uses only NOAA Coral Reef Watch variables, which are updated continuously and available globally, it can be applied without combining multiple data sources with different coverage periods. This is what allowed the 2024 validation to be carried out at all.

4.3 Comparison With Previous Work

This study followed the approach used by Arcodia et al. (2025) for Florida's Coral Reef, applying it to a region with different thermal conditions and far less monitoring data. The general result was similar: a tree-based model using satellite thermal variables can forecast heat stress onset several weeks in advance, and outperforms both linear and frequency-based baselines.

The feature importance analysis showed a difference, however. In the Florida study, the week of the year became the dominant predictor at longer lead times, suggesting the model relied on seasonal timing when direct thermal signals were still weak. In this study, HotSpot took that role instead, ranking above DHW for distant-risk predictions.

A possible explanation lies in the seasonal structure of the two regions. In the central Red Sea, thermal stress is confined to a narrow and highly regular window, with DHW at zero from February through May in every study year and a consistent peak in October. When the seasonal pattern is this sharp, the current instantaneous heat signal may carry more information about what follows than the calendar date does. This interpretation is consistent with the data but was not tested directly, and comparing feature importance across more regions would be needed to confirm it.

4.4 Limitations

The model performed poorly on the distant-risk category, with an F1 score of 0.125 for the baseline model and zero for the extended model. The cause is visible in the data: only 57 training days fell into this category, compared with 1,719 no-risk days. Class weighting reduced but did not remove this problem. In practice, the model reliably separates urgent risk from no risk, but should not be relied on for lead times beyond two weeks. This category also showed an unexpected pattern: logistic regression outperformed XGBoost here (0.241 versus 0.125), which suggests the tree-based model overfits when training examples are this scarce.

The exact onset date of bleaching is not reported in either source study. Both describe surveys carried out in a specific month, not the day bleaching began. Lead times reported here are therefore measured against the survey periods and against the DHW threshold, not against bleaching onset itself.

The solar radiation dataset ends in December 2022, which limited the extended model to the 2004–2022 period and prevented its application to the 2024 event. Only the baseline model could be tested externally.

Finally, this study used a single virtual station representing one coastal region. Bleaching severity varied between sites in both documented events, and a station-level model cannot capture that variation.

4.5 Future Work

Three directions follow from these results. The first is extending the approach to additional stations along the Saudi coast, which would allow site-level differences in thermal response to be examined. The second is testing supplementary predictors that operate on timescales closer to the prediction window, since the seasonal and climatological variables tested here did not improve performance. The third is applying the model prospectively rather than retrospectively, by running it on current-season data and recording whether its warnings are followed by observed bleaching.

5. Conclusion

This study built a machine learning model to predict coral bleaching risk in the central Saudi Arabian Red Sea. Bleaching events in this region have become more frequent and more severe, but long-term monitoring data are still limited.

An XGBoost model was trained on satellite thermal variables from NOAA Coral Reef Watch. On the 2019–2022 test data, it separated urgent risk from no risk with an F1 score of 0.972, outperforming both a linear model and a calendar-based model. The model was then applied to two documented bleaching events. In 2020, the first warning came ten days before DHW crossed the bleaching threshold. In 2024, a season that fell outside both the training and testing periods, the result was the same: a warning ten days ahead, issued at almost the same DHW value and confidence level. Since 2024 was never seen by the model, this repetition suggests it learned a stable relationship rather than patterns tied to one period.

Adding historical thermal variability, the Dipole Mode Index, and cloud ratio did not improve performance. The three thermal variables from a single continuously updated source were enough, which also keeps the model simple to apply.

The framework has clear limits. It does not predict reliably beyond two weeks ahead, because too few training examples were available for longer lead times. It represents one coastal region rather than individual reef sites. Its lead times are measured against survey periods and the DHW threshold, not against the day bleaching began, since that date is not reported in the source studies.

Within these limits, the results show that a localized model can be built for the Red Sea using data that are already public, alongside the global outlooks already running. As bleaching events intensify in this region, tools like this may help direct monitoring and protection to the sites and times where they matter most.

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